

LeafCarbon: low-cost, non-destructive estimation of plant carbon from images

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0. Executive Summary

For biological carbon removal (afforestation, soil, restoration, microalgae), the main difficulty is usually measurement: determining how much carbon was captured, accurately and at low cost. This is the MRV step (Monitoring, Reporting, Verification). The current approach uses destructive sampling, oven-drying, weighing, and laboratory elemental analysis, performed one sample at a time, which is not economic for a smallholder or a large area. As a result, an operator may be capturing carbon and still be unable to earn credits, because the amount cannot be measured affordably and its permanence cannot be demonstrated.

LeafCarbon is a low-cost, non-destructive method for this measurement. A standardized low-cost imaging rig and an AI model convert a photo and the local environment record into an estimate of a plant's biomass, and from that its carbon (carbon = dry biomass × the near-constant 0.47 fraction). A small set of oven-dried dry weights anchors the model to ground truth, and every output reports a confidence interval, so only the uncertain samples are sent to the lab. No combustion or destruction is involved.

We have built the system and measured its accuracy. On our own day-by-day lettuce dataset (569 photos), the per-plant biomass model reaches $R^2 \approx 0.64$ under leave-one-day-out cross-validation, using image structure features and environmental history. This is a measured result, not a simulation. We use lettuce because its roughly 35-day cycle allows fast iteration and lets us validate the full workflow; the same setup is intended to transfer to durable carbon pools.

LeafCarbon is not a carbon-credit issuer, and we make no claim that lettuce sequesters carbon. The aim is to provide the measurement that capture projects, smallholders, and verifiers currently lack.

1. The problem and technical value

Carbon-market credibility depends on measurement, and measurement is the main bottleneck. Obtaining a plant's biomass requires harvesting it, drying it to constant weight, and weighing it; obtaining its carbon content requires laboratory elemental analysis (dry combustion / CHN). Both destroy the sample and cannot be repeated on the same plant, and each has a real cost: laboratory soil-carbon analysis is about US\$20–75 per sample (US\$20.77 for one thermal-combustion run,

higher once field sampling and preparation are included) [8]. Across fragmented smallholders or large areas, this does not scale.

The bottleneck has financial consequences. Operators who capture carbon often cannot sell credits, because the amount captured is hard to measure affordably and its permanence is hard to prove. The limiting factor is the cost of measurement rather than the capture itself.

A standard for a credible method already exists, and so does a gap. Taiwan's Forestry Research Institute published an official, MRV-compliant carbon methodology in 2025 [14], with third-party verification and sampling uncertainty held below 10%. It covers forests only. For the other pools the gap is not a missing methodology but the cost of the measurement each one demands: international voluntary methodologies already exist for soil carbon (Verra VM0042 [9]) and biochar (Verra VM0044 [17]), yet they rely on laboratory soil sampling or coarse-resolution satellite remote sensing, neither of which is economical for single plants, smallholders, or frequent monitoring. LeafCarbon is aimed at that standard and at that cost gap.

Taiwan's policy depends on the same measurement. The carbon fee is NT\$300 per t CO₂e from 2025 [15], and the agriculture sector targets net zero by 2040 with +10 Mt CO₂e of natural sinks, of which soil carbon is the largest component [16]. Both require measurement that is currently too expensive to run at scale.

Image-based MRV for biological carbon has received comparatively little development. Recent finalist cohorts in this contest are concentrated in hydrogen, CO₂ conversion, batteries, and recycled materials, so low-cost imaging-based measurement of biological carbon remains relatively unexplored.

2. Related work, and how LeafCarbon differs

The underlying science is well established, which keeps the technical risk low. The link between leaf greenness, canopy structure, and biomass dates back decades: Tucker (1979) showed that the red/near-infrared combination behind NDVI is proportional to green leaf biomass [1], and chlorophyll proxies such as the SPAD meter use the same principle. More recent work applies this to ordinary cameras: smartphone RGB images with a CNN estimate pearl-millet biomass non-destructively [2]; structure and texture features feed machine-learning biomass models across crops [4]; and, most relevant here, RGB-depth imaging with morphometric descriptors and machine learning estimates biomass in hydroponic lettuce [3]. We do not claim to have invented image-to-biomass regression; we build on it.

What is comparatively undeveloped is a measurement instrument designed for carbon projects, and the existing companies operate at a different scale. The main digital-MRV providers measure soil carbon from satellites. Boomitra combines imagery from more than 100 satellites with an archive of over a million laboratory-analyzed soil samples, is the first remote-sensing method approved under Verra's VM0042 soil-carbon methodology, won the 2023 Earthshot Prize, and reports reducing MRV cost by more than 90% relative to traditional sampling [9]. CarbonFarm and Treeconomy use similar

satellite-based approaches for rice and forests [13]. These tools are large-area, single-pool, and tied to credit issuance, and satellite methods are limited for below-ground and fine-grained carbon [10].

LeafCarbon addresses a complementary case: ground-level, single-plant, non-destructive optical measurement that issues no credits. Satellite dMRV covers large areas at coarse resolution, while LeafCarbon measures individual plants with a low-cost camera and reports a per-plant confidence interval. It is an upstream measurement module that a capture project can adopt, rather than a competing platform or a methodology owner.

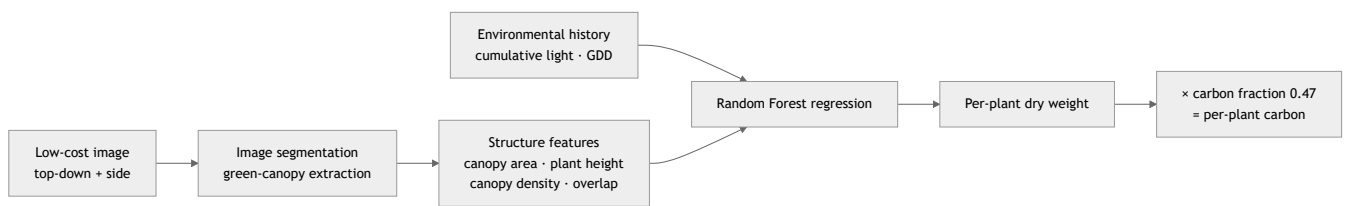
3. The measurement setup

LeafCarbon is a deployable measurement kit, consisting of a capture rig, an AI estimator, and a calibration anchor:

Component	Content
Standardized capture	Fixed-height photography (top-down and side) under consistent lighting: a simple, low-cost rig that others can reproduce.
AI estimator	Image → plant structure features (canopy area, plant height, canopy density) plus environmental history → per-plant dry biomass, then × carbon fraction → carbon.
Ground-truth calibration	A small set of samples with oven-dried dry weight as the calibration anchor; carbon = dry weight × the near-constant 0.47 fraction. No elemental analysis or combustion needed.
Uncertainty estimate	Every estimate carries an uncertainty range, which carbon verification (MRV) needs; only the least certain samples are sent to the lab.

4. The AI model

The full pipeline, from image to per-plant carbon:



The model uses interpretable structure features throughout. Calibration needs only a small set of oven-dried dry weights as ground truth; carbon = dry weight × 0.47, with no elemental analysis or combustion.

4.1 How the structure features are computed (formulas)

Every structure feature is computed geometrically from the green-canopy mask, so each is traceable and reproducible:

Step 1: green-canopy extraction. The image is converted to HSV; we take the green hue band (hue $H \in [33^\circ, 90^\circ]$, saturation $S \geq 45$, value $V \geq 35$) and additionally require a positive Excess-Green index:

$$\text{ExG} = (2G - R - B) / (R + G + B) > 0.02$$

The intersection of both conditions, followed by a morphological opening to remove stray specks, yields the canopy mask; the green-pixel count is A_{green} . The dual condition keeps the segmentation stable even when green is a small fraction of the frame, rejecting brown soil, grey background and the white rig (a plain Otsu threshold mis-segments background as green when green is scarce).

Step 2: geometric features from the mask. From the mask's bounding box (width w , height h ; image width W , height H) and convex-hull area A_{hull} :

Structure feature	Formula	Meaning
height	h / H	in a side view, bbox height tracks plant height
aspect	w / h	spreading vs. upright shape
green cover	$A_{\text{green}} / (W \cdot H)$	overall proxy for leaf amount
extent	$A_{\text{green}} / (w \cdot h)$	how fully the canopy fills its bbox
solidity	$A_{\text{green}} / A_{\text{hull}}$	proxy for leaf overlap / compactness

Step 3: aggregate per plant. Per-plant values are the mean over the side-view angles (the top-down view is kept separate), reducing single-viewpoint bias. These structure features, together with cumulative environment (temperature, cumulative sun, GDD, cumulative UV), are the inputs to the Random Forest, the same values shown in the "model inputs" panel of the online demo.

5. Carbon logic and technical contribution

Estimating biomass from leaf greenness and structure is an established field [1], [2], [3], [4]; NDVI and the SPAD chlorophyll meter have demonstrated that the approach works, so our effort goes into integration and deployment rather than into the underlying regression.

Our contribution is the measurement tool and its configuration: a low-cost, non-destructive carbon-measurement tool designed for carbon projects and validated end to end on data we collected.

The carbon logic is straightforward. The carbon fraction of plant dry matter is approximately constant (about 47%; IPCC 2006 default 0.47 tC/t [5]; Ma et al. 2018 report 45.0–47.9% across 4,318 species [6]). The quantity worth measuring is therefore biomass, with carbon = biomass $\times$ 0.47. The image estimates biomass; the small set of oven-dried weights calibrates the image-to-biomass relationship; carbon is then obtained from the 0.47 constant, with no elemental analysis. Lettuce is a fast validation crop: its roughly 35-day cycle lets us tune the image-to-biomass relationship and

validate the workflow quickly, on a setup designed to transfer to durable pools such as forests, soil, and biochar.

This proxy approach is the one the official standard already uses. Taiwan's national forest-carbon method does not harvest trees; it measures trunk diameter and height and applies published coefficients (volume $\rightarrow$ biomass $\rightarrow$ carbon $\rightarrow \times 44/12$), accepting a model only when it reaches $R^2 \geq 0.85$ and keeping overall sampling uncertainty below 10% under third-party verification [14]. LeafCarbon follows the same observable $\rightarrow$ model $\rightarrow$ biomass $\rightarrow$ carbon sequence, using canopy images in place of trunk calipers, which lets it cover the small plants, smallholders, and frequent monitoring that a trunk-based method cannot.

6. Results and cost comparison

On the 72 single-plant samples (24 days) in our lettuce dataset that have individual oven-dried weights, the per-plant biomass model reaches $R^2 \approx 0.64$, $RMSE \approx 0.43$ g under leave-one-day-out cross-validation, using side-view structure features and cumulative environment. Leave-one-day-out cross-validation withholds an entire day the model has not seen, so the result is out-of-sample.



Fig 1: Plant canopy extracted from low-cost images across growth (days 7, 21, 33); background, floor and rig excluded.

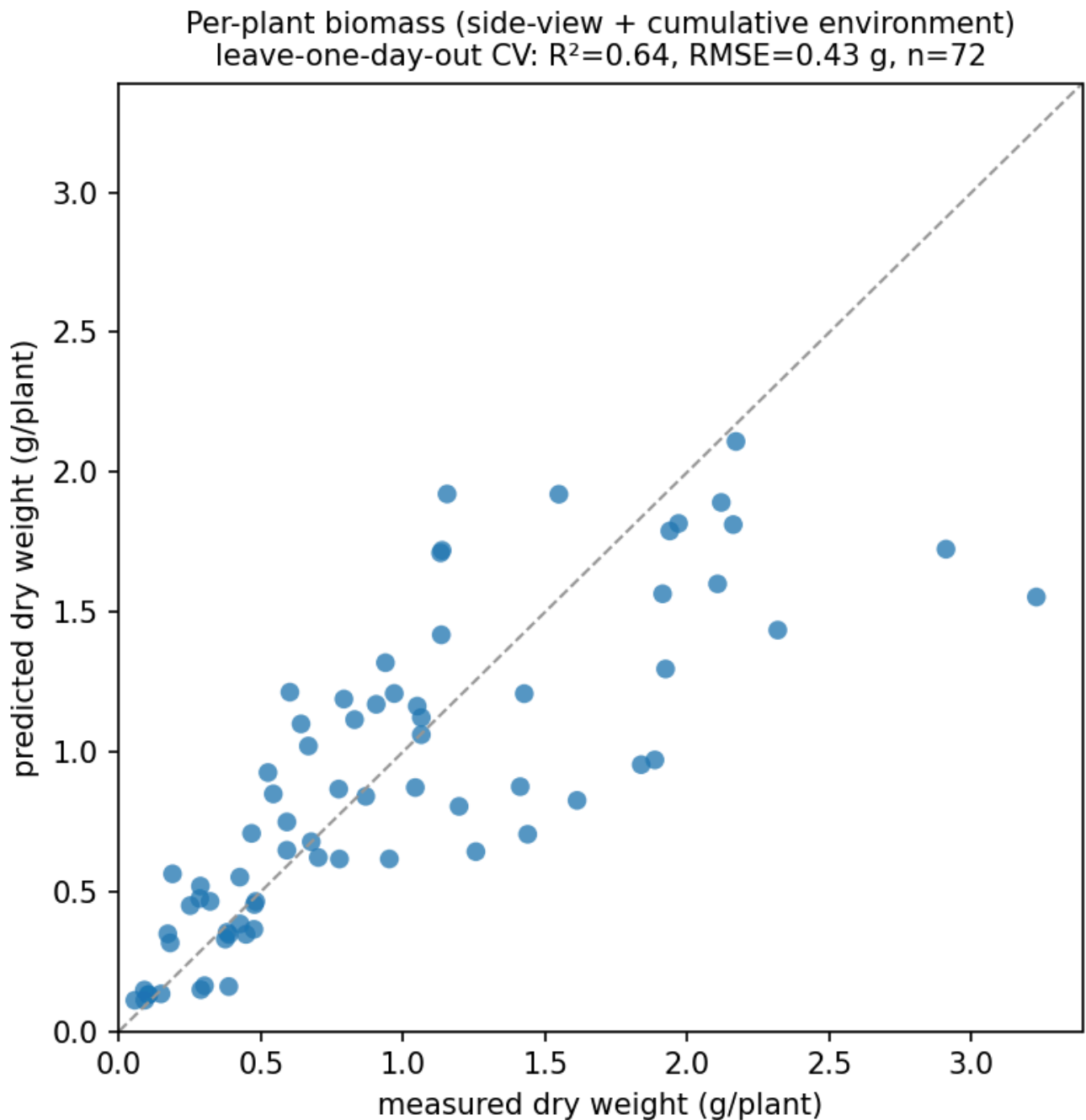
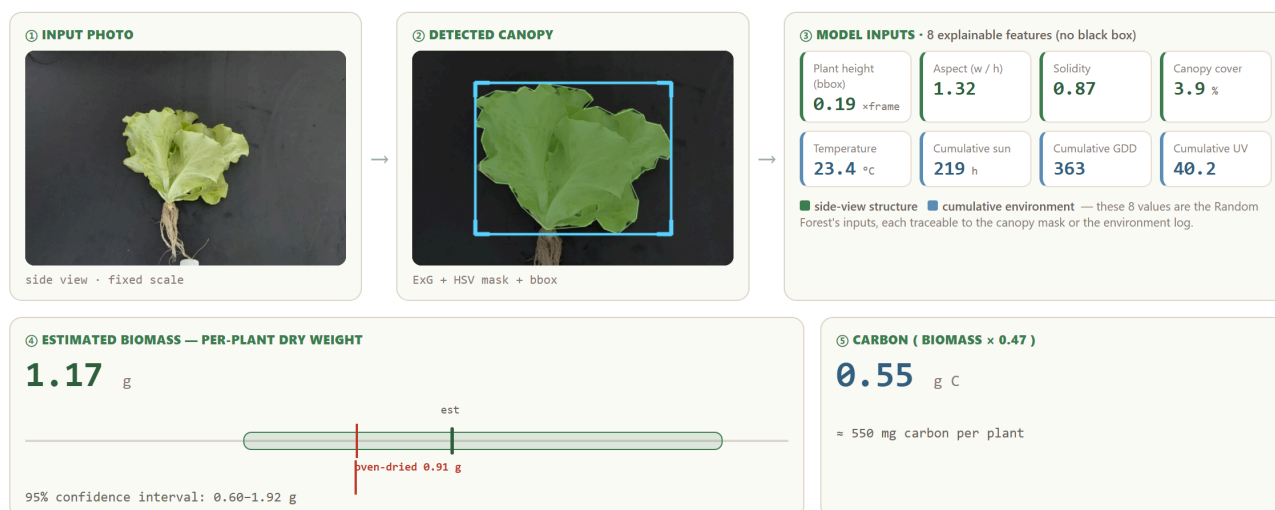


Fig 2: Predicted versus measured per-plant dry weight. Each dot is one lettuce plant: the x-axis is its measured dry weight after oven-drying (ground truth), the y-axis is the model's estimate from a single photo and the local environment record. The dashed diagonal indicates perfect prediction. Overall $R^2 \approx 0.64$ (the model accounts for about 64% of the plant-to-plant variation in dry weight), RMSE ≈ 0.43 g, under leave-one-day-out cross-validation. $n = 72$ plants.

To make the pipeline concrete, Fig 3 follows a single plant end to end: one low-cost photo, the detected canopy, the eight explainable features that go into the model, the per-plant dry-weight estimate with its 95% confidence interval, and the carbon figure. It is the static form of our interactive demo, so the same walkthrough is in the document for readers who do not open the demo.

Worked example — one plant, end to end single low-cost photo · Day 27 · Random Forest, leave-one-day-out CV



The oven-dried ground truth for this plant (**0.91 g**) falls inside the model's 95% interval: the estimate reports its own uncertainty. In operation, only plants whose interval is too wide or excludes expectation are sent to the lab, so an audited result is reached at a fraction of the destructive-sampling cost.

Fig 3: The full pipeline on one real plant (Day 27). A single side-view photo → green-canopy detection → eight explainable features (four side-view structure, four cumulative environment) → per-plant dry weight with a 95% confidence interval → carbon = dry weight × 0.47. The oven-dried ground truth for this plant (0.91 g) lies inside the predicted interval, illustrating the self-reported uncertainty that decides which plants are sent to the lab (Section 7). This is the static form of the interactive demo.

The objective of LeafCarbon is to reduce the cost per measurement, not to exceed the accuracy of existing methods. Its position relative to the alternatives:

Method	Destructive?	Per-measurement cost	Coverage	Issues credits?
Lab elemental analysis (CHN / dry combustion)	Yes	~US\$20–75 per sample [8]	Single sample, lab turnaround	No (measurement only)
Destructive harvest + oven-dry + weigh	Yes	Labour per plant, not repeatable	Single plant	No
Satellite dMRV (e.g. Boomitra)	No	>90% cheaper than traditional sampling [9]	Large area, coarse, below-ground is hard [10]	Yes (Verra VM0042)
LeafCarbon	No	A photo (calibration set built once); near-zero marginal cost per added plant	Single plant, ground level, repeatable	No (upstream measurement module)

LeafCarbon and satellite dMRV are complementary rather than competing: satellite methods cover large areas, and LeafCarbon resolves individual plants. Relative to the laboratory route, the saving is

structural, because once the calibration set is built, each additional plant requires only a photo.

7. Limitations and scientific positioning

This section states what LeafCarbon does and does not claim.

We make no sequestration claim for lettuce. An annual herbaceous crop is not permanent carbon removal: on average only about 7% of the carbon a crop assimilates remains in the soil after harvest, while the rest returns to the atmosphere, so the net carbon stock change is close to zero and standard accounting does not credit it [7]. Lettuce is a model organism here, used to tune the instrument rather than to claim a sink.

Measurement is distinct from a carbon credit. Measuring how much carbon a plant holds is a technical task; issuing a tradable credit requires an approved methodology and proof of permanence. LeafCarbon performs the measurement and does not issue credits, which keeps it separate from the permanence requirement that constrains many biological-carbon proposals.

Current accuracy is below the audited threshold. The forest standard requires sampling uncertainty below 10% and a model at $R^2 \geq 0.85$ [14]; our $R^2 \approx 0.64$ does not yet meet this. For that reason every output carries a confidence interval and only the uncertain plants are sent to the lab, so an audited result can reach the below-10% target at lower cost.

The most difficult step is the conversion from fresh weight to dry weight. Biomass and water content account for most of the variation, and this conversion is where RGB imaging is weakest. Near-infrared reflectance measures water and dry matter more directly, so a handheld NIR channel is a planned accuracy improvement.

Transfer to other pools is an objective, not a demonstrated result. The setup is designed to extend to durable pools (forest, soil, biochar), but each pool requires its own calibration set and validation. We claim a validated workflow on lettuce and a documented path toward those pools, not that the lettuce results apply directly.

8. Expected benefit and roadmap

The benefit is in cost and speed. The conventional method for obtaining one plant's biomass and carbon is destructive harvest, oven-drying, weighing, and laboratory elemental analysis, performed sample by sample and not repeatable. LeafCarbon estimates these from a photo within seconds, without destroying the plant, and the same plant can be measured again later, at near-zero marginal cost per additional plant. At project scale, digital MRV has been reported to reduce per-farm MRV cost substantially (Boomitra reports over 90% relative to traditional sampling [9]; the wider digital-MRV literature reports comparable reductions [10]).

For AI performance, the $R^2 \approx 0.64$ / $RMSE \approx 0.43$ g figures are a measured leave-one-day-out result, with the model fully specified (Section 6).

For market context, lower-cost verification allows afforestation, soil-carbon, and restoration projects, and smallholders, to participate; the voluntary agricultural carbon market is projected to grow at roughly 32% CAGR through 2034 [12]. The market is sensitive to quality and contracted in 2024 on credibility concerns [11], which indicates demand for transparent measurement with reported uncertainty. The approach also aligns with Taiwan's carbon fee [15] and the 2040 agricultural-sink target [16].

The roadmap: (1) validate image-to-biomass on the lettuce dataset, completed at $R^2 \approx 0.64$; (2) build a biomass and carbon gradient from batch-harvested lettuce; (3) anchor it with a small set of oven-dried weights, with CHN where available; (4) add a handheld NIR channel for the fresh-to-dry step; (5) pilot transfer to one durable pool with its own calibration set, as the measurement module for a capture project.

9. Team

We are a Computer Science and Materials Science student team. CS builds the AI model, the routing logic, and the system. Materials handles the ground-truth side: oven dry-weight calibration, imaging standardization, and the measurement procedure. A measurement needs two things, a model and reference values to check it against; CS does the first, Materials the second. Carbon-credit methodology, soil science, and agronomy are separate areas that belong to the projects using our data, not to the measurement tool, so we make no claim in them.

10. References

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